

# ENME331 Exam 1 Solver Guide for AI Models

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## Ground rules for the solver

Solve every problem using only the ENME331 Exam 1 formula sheet below plus basic algebra, geometry, calculus and unit conversion. Fluid properties not given in the problem may be taken from the standard values table in this guide.

**Course scope (Exam 1):** fluid statics, kinematics, Bernoulli equation. Textbook: Gerhart, Fundamentals of Fluid Mechanics (Munson), 8th ed., Chapters 1 to 3. Homework is graded on WileyPlus, which checks numeric answers to roughly 2 percent, so carry 4 significant figures until the end.

**Notation (use exactly these symbols, never underscores):**  $p$  or  $P$  pressure,  $\rho$  density,  $\gamma$  =  $\rho g$  specific weight,  $SG$  specific gravity,  $h$  depth or height,  $z$  elevation (positive up),  $V$  velocity,  $Q$  volume flowrate,  $A$  area,  $D$  or  $d$  diameter,  $R$  radius of curvature,  $h_{\text{CG}}$  centroid depth,  $y_{\text{CG}}$  centroid location along the surface,  $y_R$  written as  $yR$  center of pressure,  $I_{\text{CG}}$  or  $I_{xc}$  second moment of area about the centroid. Write subscripts as Unicode or plain letters ( $P_1$ ,  $\gamma_2$ ,  $h_c$ ), not with underscores.

### Required answer format for every problem:

1. One or two sentences naming the physical concept and why it applies.
2. A "Given" list with every value converted to consistent units.
3. Numbered steps, each with the equation first, then numbers substituted, then the result with units.
4. The final answer in bold with units and the sign convention stated (gauge vs absolute, direction of force, which side of a point).
5. A one-line sanity check (order of magnitude, sign, physical intuition).

**Unit systems:** never mix them. In US units use lb, ft, s, slug;  $\gamma_{\text{water}} = 62.4 \text{ lb/ft}^3$ ,  $p_{\text{water}} = 1.94 \text{ slug/ft}^3$ ,  $g = 32.2 \text{ ft/s}^2$ ,  $1 \text{ psi} = 144 \text{ lb/ft}^2$ . In SI use N, m, s, kg;  $\gamma_{\text{water}} = 9810 \text{ N/m}^3$  (or  $9.80 \text{ kN/m}^3$  if the problem says so),  $p_{\text{water}} = 1000 \text{ kg/m}^3$ ,  $g = 9.81 \text{ m/s}^2$ . Convert inches to feet and mm to m before substituting.

## Formula sheet (the only equations allowed)

Everything below is copied from the official ENME331 Exam 1 equation sheet. Every solution must trace back to these lines.

### Fluid statics: pressure

$$\frac{\partial p}{\partial z} = -\rho g \quad \int_{p_1}^{p_2} dp = -\rho g \int_{z_1}^{z_2} dz$$

Constant density:

$$p_1 - p_2 = \rho g (z_2 - z_1) \quad p_1 = p_2 + \gamma h \quad \gamma = \rho g$$

### Fluid statics: force on a flat surface

$$F_R = \gamma h_C A \quad y_R = \frac{I_{xc}}{y_C A} + y_C$$

$h_C$  is the vertical depth of the centroid below the free surface.  $y_C$  and  $y_R$  are measured along the plane of the surface from where that plane meets the free surface (for a vertical plate they equal depths).

### Buoyancy

$$F_B = \gamma_f V_s$$

$\gamma_f$  is the specific weight of the surrounding fluid,  $V_s$  the submerged volume.  $F_B$  acts through the centroid of the displaced volume.

### Section properties on the sheet (A, centroid, $I_{xc}$ )

| Shape                     | Area A      | Centroid location                     | $I_{xc}$              |
|---------------------------|-------------|---------------------------------------|-----------------------|
| Rectangle b wide, a tall  | ba          | a/2 from edge                         | ba <sup>3</sup> /12   |
| Circle radius R           | $\pi R^2$   | center                                | $\pi R^4/4$           |
| Semicircle radius R       | $\pi R^2/2$ | 4R/(3 $\pi$ ) from flat edge          | 0.1098R <sup>4</sup>  |
| Quarter circle radius R   | $\pi R^2/4$ | 4R/(3 $\pi$ ) from each straight edge | 0.05488R <sup>4</sup> |
| Triangle base b, height a | ab/2        | a/3 from base, (b+d)/3 horizontally   | ba <sup>3</sup> /36   |

## Kinematics

$$\vec{a} = \frac{\partial \vec{V}}{\partial t} + u \frac{\partial \vec{V}}{\partial x} + v \frac{\partial \vec{V}}{\partial y} + w \frac{\partial \vec{V}}{\partial z}$$

$$\text{Streamline: } \frac{dy}{dx} = \frac{v}{u} \quad \text{Pathline: } \frac{dx}{dt} = u, \quad \frac{dy}{dt} = v$$

## Steady, incompressible, inviscid flow

Along a streamline (Bernoulli):

$$\frac{1}{2} \rho V^2 + p + \rho g z = \text{constant}$$

Normal to a streamline (n points toward the center of curvature):

$$\rho \frac{V^2}{R} = -\frac{\partial p}{\partial n} - \rho g \frac{dz}{dn}$$

Conservation of mass:

$$Q = AV = \text{constant} \quad \text{or} \quad \dot{m} = \rho A V$$

Static pressure  $p$ , dynamic pressure  $\frac{1}{2}\rho V^2$ , stagnation pressure  $p + \frac{1}{2}\rho V^2$ .

## Fluid statics: pressure methods

In a fluid at rest, pressure depends only on elevation. Every statics pressure problem is solved by walking from a point of known pressure to the target point.

**The walking rule (from  $p_1 = p_2 + \gamma h$ ):**

- Going DOWN a distance  $h$  through fluid of specific weight  $\gamma$ : add  $\gamma h$ .
- Going UP: subtract  $\gamma h$ .
- Moving horizontally within the same connected fluid: no change.
- Crossing an interface between two fluids: pressure is equal on both sides.
- Gas columns (air) in manometer tubes: neglect their weight unless told otherwise.

## Manometers

1. Label every interface and every point where a tube changes fluid.
2. Start at an open end ( $p = 0$  gauge) or a known pressure.

3. Walk point to point, using the  $\gamma$  of whichever fluid fills that stretch.
4. For a differential U-tube between two points at the same elevation in a flowing or static fluid of  $\gamma$ , with gage fluid  $\gamma_{gf}$  and reading  $h$ :  $p_1 - p_2 = (\gamma_{gf} - \gamma)h$ . The side where the gage fluid is pushed down has the higher pressure.
5. Inverted U-tube: the lighter fluid sits on top; same walking rule, just track which legs hold which fluid. Read each dimension's leader line carefully: stacked dimensions often belong to different legs.

### Barometers and altitude

A mercury column reading is a pressure:  $p = \gamma_{Hg} \cdot h$ . To find a new reading after an elevation change  $\Delta z$  in air, convert reading to pressure, subtract  $\gamma_{air} \cdot \Delta z$  (constant-density approximation is acceptable over hundreds of feet), and convert back with  $h = p/\gamma_{Hg}$ .

### Pressure head

"Pressure head in mm of mercury" means  $h = p/\gamma_{Hg}$ . Use  $SG_{Hg} = 13.6$  unless the problem gives  $\gamma_{Hg}$  (for example  $847 \text{ lb/ft}^3$ ).

### Gauge vs absolute

Gauge pressure is relative to the atmosphere; absolute = gauge +  $p_{atm}$  (14.7 psia or 101.3 kPa unless given). Use gauge for forces on walls when the atmosphere acts on both sides. Use absolute whenever vapor pressure, ideal-gas law or "psia" appears. Always label which one the final answer is.

### Gases with varying density (from $\partial p/\partial z = -\rho g$ with $\rho = p/RT$ )

- Isothermal layer:  $p_2 = p_1 \cdot \exp[-g(z_2 - z_1)/(RT_0)]$
- Linear temperature  $T = T_0 - \beta(z - z_0)$ :  $p = p_0[1 - \beta(z - z_0)/T_0]^{(g/(R\beta))}$

Use  $R = 287 \text{ J/(kg}\cdot\text{K)}$  for air, temperatures in kelvin. These come from integrating the sheet's first equation, so they are allowed.

## Forces on surfaces, gates and buoyancy

Hydrostatic force always acts perpendicular to the surface. Use gauge pressure when air is on the dry side.

### Flat surfaces

- Magnitude  $F_R = \gamma h_C \cdot A$ , where  $h_C$  is the vertical depth of the wetted area's centroid.

- Location  $y_R = I_{xc}/(\gamma C \cdot A) + y_C$ , measured along the plate from the line where the plate's plane meets the free surface.  $y_R$  is always below the centroid.
- Shortcut: a rectangle whose wetted edge starts at the free surface has a triangular pressure load;  $F_R$  acts  $1/3$  of the wetted length up from the deepest edge.
- Only the wetted portion counts. If a plate is longer than the water depth,  $A$  and  $h_C$  use the submerged length only.
- Inclined plate at angle  $\theta$  to horizontal: vertical depth = (distance along plate)  $\cdot \sin\theta$ .

**Pressurized tanks:** if the free surface has gauge pressure  $p_0$ , replace it with an imaginary extra water depth  $p_0/\gamma$  above the real surface, then use the formulas.

### Curved surfaces

- Horizontal component  $F_H$  = force on the vertical projection of the curved surface (a flat rectangle), using  $F_R = \gamma h_C \cdot A$  with that projection's centroid depth. Check what depth  $h$  is measured to on the figure: the top of the curve, its center, or its bottom.
- Vertical component  $F_V$  = weight of fluid directly above the curved surface up to the free surface (real or imaginary). For a full semicircular bulge on a vertical wall, the above and below volumes cancel except the bulge itself:  $F_V = \gamma(\pi R^2/2) \cdot L$ , downward.

### Gates, hinges, stops: always a moment balance

1. Draw a free body diagram of the gate only: hydrostatic forces, gate weight at its centroid, cable or stop forces, hinge reactions.
2. Sum moments about the hinge so the hinge reactions drop out.
3. Moment arm for a force perpendicular to a straight gate = distance along the gate from the hinge. Weight (vertical) uses the horizontal distance. A horizontal cable uses the vertical distance.
4. "About to open" or "no force on the stop" means set the stop force to zero.
5. Forces that pass through the hinge produce no moment (for example pressure on a circular arc centered on the pivot).
6. Right-angle or Z-shaped gates: treat each segment separately, then add moments. A horizontal segment resting on a solid floor only feels an unbalanced force over an opening (a drain pipe):  $F = \gamma h \cdot A(\text{opening})$  at the opening's center.
7. Hinge location sets every horizontal-force moment arm. Confirm whether the hinge is at the free surface, above it, or below it before integrating.

For segments with linearly varying pressure, integrate directly:  $M = \int (\text{arm}) \cdot \gamma(\text{depth}) \cdot L \cdot ds$ .

### Buoyancy and submerged bodies

- $FB = \gamma_{\text{fluid}} \cdot V_{\text{submerged}}$ , acting at the centroid of the displaced volume.
- For composite bodies, list each part's weight  $W = \gamma_{\text{part}} \cdot V$  (or  $SG \cdot \gamma_{\text{water}} \cdot V$ ) and buoyancy separately with its own centroid.
- Force balance gives the holding force; moment balance about a point gives its location:  
 $x = \Sigma(\text{net upward force} \times x)/F$ .
- Triangle centroid = average of its three vertices.

## Kinematics

A velocity field  $V = (u, v, w)$  gives the velocity at every point and time. Steady means no explicit  $t$  in  $u, v, w$ ; otherwise unsteady.

**Speed:**  $|V| = \sqrt{u^2 + v^2 + w^2}$ . Simplify algebraically before plugging in; terms often cancel (for example  $u = -V_0 y / \sqrt{x^2 + y^2}$ ,  $v = V_0 x / \sqrt{x^2 + y^2}$  gives speed  $V_0$  everywhere except the origin).

**Streamlines (a snapshot at fixed  $t$ ):**

1. Write  $dy/dx = v/u$ .
2. Treat  $t$  as a constant.
3. Separate variables and integrate.
4. Apply the point the streamline passes through to find the constant.
5. Sketch or match to curves by checking one or two points (for example  $y^2 = 2tx$  at  $x = 16$  gives  $y = 5.7$  for  $t = 1$ ,  $y = 8$  for  $t = 2$ ).

**Pathlines (where a particle actually goes):** solve  $dx/dt = u$  and  $dy/dt = v$  together with the particle's starting position at its starting time.

**Streamlines vs pathlines:** they coincide in steady flow. In unsteady flow they generally differ. A point where  $u = v = 0$  for all  $t$  (a stagnation point) has a pathline that is just that point.

**Acceleration:** apply the sheet formula component by component. The  $\partial/\partial t$  term is the local (unsteady) part; the  $u\partial/\partial x + v\partial/\partial y + w\partial/\partial z$  terms are convective. A steady flow can still accelerate through convection.

**Free vortex recognition:** circular streamlines  $x^2 + y^2 = C$  come from  $dy/dx = -x/y$ .

## Bernoulli and continuity

Bernoulli applies along a streamline when the flow is steady, incompressible, inviscid, and has no pump or turbine (no mechanical work). Check all five before using it; for a checklist question, answer each one explicitly.

### Standard simplifications

- Large tank surface:  $V \approx 0$ .
- Open tank surface or free jet exit:  $p = 0$  gauge. A free jet's exit pressure equals the surrounding pressure across the whole exit plane.
- Same elevation:  $z$  terms cancel.
- Head form (divide by  $\gamma$ ):  $p/\gamma + V^2/2g + z = \text{constant}$ . Each term is a height.

**Continuity:**  $A_1 V_1 = A_2 V_2$ , so  $V_2 = V_1 (D_1/D_2)^2$ . Always pair it with Bernoulli when two diameters appear.

### Free jets and Torricelli

- Tank draining through an exit  $h$  below the surface:  $V = \sqrt{2gh}$ .
- A vertical jet rising height  $h$  above the nozzle:  $V(\text{exit}) = \sqrt{2gh}$ .
- Pressurized tank, nozzle at the surface level:  $p(\text{tank}) = \gamma h(\text{rise})$ .

**Pitot (stagnation) tubes:** a tube opening facing into the flow brings fluid to rest, so it reads stagnation pressure  $p + \frac{1}{2}\rho V^2$ . A static tap on the wall reads  $p$ . A manometer between a static tap and a pitot tube reads  $\frac{1}{2}\rho V^2$  directly. Identify each manometer leg as static or stagnation from the figure before writing equations; a tube poking into a free jet is a pitot tube.

**Venturi and contraction meters:** two static taps at different diameters, same elevation:  $p_1 - p_2 = \frac{1}{2}\rho(V_2^2 - V_1^2)$  with  $V_1 = V_2(A_2/A_1)$ . Combine with the manometer equation.

**Cavitation:** begins where absolute pressure falls to vapor pressure  $p_v$ , which is the narrowest, fastest, highest section. Work in absolute pressure. To avoid it: lower the velocity there (widen the throat, shrink the exit that sets flowrate, lower the driving head).

### Pressure across curved streamlines (normal direction)

- Horizontal flow:  $dp/dr = \rho V^2/r$ , pressure increases outward from the center of curvature.
- In a pipe bend, the inside wall has lower pressure than the outside wall.
- Straight parallel streamlines ( $R \rightarrow \infty$ ): pressure varies only hydrostatically across them.

- For air, use  $\rho = 0.00238 \text{ slug/ft}^3$  or  $1.2 \text{ kg/m}^3$  unless given; convert mph to ft/s with  $\times 5280/3600$ .

**Comparison questions (true, false, could be either):** compare by asking which effects push each pressure up or down. If two effects push in opposite directions and their sizes aren't fixed, the answer is "could be either."

## Traps that caused wrong answers before

Each of these produced a graded mistake on Sam's homework. Check every one before finalizing.

| Trap                               | What went wrong                                                                                                      | Rule                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Depth reference on curved surfaces | Used $h$ as the bulge's center depth; $h$ was to the top of the bulge (FH 535 lb wrong, 746 lb right)                | Read exactly where every dimension arrow starts and ends                                  |
| Manometer leg type                 | Treated a tube inside a free jet as a static tap; it was a pitot tube (Q 0.346 wrong, 5.52 ft <sup>3</sup> /s right) | Classify each leg as static or stagnation first                                           |
| Hinge location                     | Assumed hinge A was $h$ above the surface; the key placed it at the surface ( $b = 9h$ vs $\sqrt{54}h$ )             | Confirm hinge elevation relative to the free surface; it sets every horizontal moment arm |
| Stacked dimensions                 | Read three stacked labels as one leg's total height                                                                  | Follow each leader line to its own leg; one measurement per unknown                       |
| Gauge vs absolute                  | Answer box expected one, solution gave the other                                                                     | State which, and give both if unsure                                                      |
| Mixed units                        | Using 847 lb/ft <sup>3</sup> in an SI problem                                                                        | Convert or use $SG \times \gamma_{\text{water}}$ in the problem's own units               |
| Wetted area                        | Using the full gate length instead of the submerged length                                                           | Only wetted area carries pressure                                                         |

| Trap               | What went wrong                                                                                            | Rule                                     |
|--------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Sign of $\Delta P$ | Confusion over $p(\text{top}) = p(\text{base}) - \gamma h$ vs $p(\text{top}) - p(\text{base}) = -\gamma h$ | Define $\Delta P$ once and keep its sign |

**Before submitting any answer, verify:**

- ☐ Every dimension used is read from its own arrows on the figure
- ☐ Units consistent throughout
- ☐ Gauge or absolute stated
- ☐ Direction of each force or the side of each point stated
- ☐ Magnitude passes a sanity check

When the figure is ambiguous, state the interpretation used and give the alternate answer in one line.

## Worked example bank

These are solved problems from this course with confirmed or checked answers. Use them as templates and as calibration.

| # | Problem type                                                                                                                       | Key equations                                                                      | Answer                                                                   |
|---|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 1 | Barometer reads 29.97 in Hg at base; carried up 500 ft ( $\gamma_{\text{air}} 0.0765$ , $\gamma_{\text{Hg}} 847 \text{ lb/ft}^3$ ) | $p = \gamma_{\text{Hg}} \cdot h$ ; $\Delta p = \gamma_{\text{air}} \cdot \Delta z$ | 29.43 in Hg                                                              |
| 2 | Closed tank, $p_A = 60 \text{ kPa}$ , piping with SG 0.8 liquid: $p$ in pipe B, head at dome top C                                 | Walking rule; $h = p/\gamma_{\text{Hg}}$                                           | $p_B \approx 103 \text{ kPa}$ ; 230 mm Hg                                |
| 3 | Inverted U-tube with oil SG 0.8 between $\text{CCl}_4$ tank at 8 psi and open water tank                                           | Walking rule across three fluids                                                   | $h \approx 22.4 \text{ ft}$ (check leg assignment of stacked dimensions) |

| #  | Problem type                                                                                                           | Key equations                                                              | Answer                                                                              |
|----|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 4  | Closed tank, gauge 7 psi, 4 ft water: open column h, pressure on AB, absolute air pressure                             | $p = p_0 + \gamma h$                                                       | $h \approx 20.2 \text{ ft}$ ; 8.73 psi; 21.7 psia                                   |
| 5  | 4 ft wide, 11 ft gate, 800 lb, $\theta = 68^\circ$ , wetted $L_2 = 6.2 \text{ ft}$ , horizontal cable at top           | $\Sigma M$ about hinge; F at $L_2/3$                                       | $T \approx 1063 \text{ lb}$                                                         |
| 6  | Right-angle gate over 3 ft drain, $w = D = L = 3 \text{ ft}$                                                           | $\gamma h^2 w / 2 \cdot (h/3) = \gamma h (\pi D^2 / 4) L$                  | $h \approx 6.51 \text{ ft}$                                                         |
| 7  | Semicircular bulge $R = 1.3 \text{ ft}$ , $h = 3.3 \text{ ft}$ to the bulge top, 1 ft length                           | $FH = \gamma(h+R)(2R)$ ; $FV = \gamma \pi R^2 / 2$                         | $FH \approx 746 \text{ lb}$ ; $FV \approx 166 \text{ lb}$                           |
| 8  | Wood (SG 0.6) triangle on aluminum plate (168 lb/ft <sup>3</sup> ) in oil SG 0.8                                       | $\Sigma F$ and $\Sigma M$ with buoyancy                                    | $F \approx 682 \text{ lb up}$ ; 6.22 ft right of A                                  |
| 9  | $u = -V_0 y/r$ , $v = V_0 x/r$                                                                                         | Speed; $dy/dx = v/u$                                                       | Speed $V_0$ everywhere; $x^2 + y^2 = C$                                             |
| 10 | $V = x^2 y \mathbf{i} + x^2 t \mathbf{j}$ , streamlines through origin                                                 | $y^2 = 2tx$                                                                | $t=0$ : curve 1, $t=1$ : curve 3, $t=2$ : curve 4; no, particle at origin stays put |
| 11 | Pressurized tank, jet rises 16 ft, nozzle level with surface                                                           | $p = \gamma h$                                                             | 6.93 psi gauge                                                                      |
| 12 | Closed tank 8 ft above pipe, 1 ft pipe to 3 in exit, Hg manometer 0.73 in between static tap and pitot tube in the jet | $\frac{1}{2} \rho V_1^2 = (\gamma Hg - \gamma) h$ ; Bernoulli tank to exit | $Q \approx 5.52 \text{ ft}^3/\text{s}$ ; $p(\text{air}) \approx 81.6 \text{ psi}$   |
| 13 | Open tank $h_1 = 3.8 \text{ m}$ , $d_3 = 0.08 \text{ m}$ , exit $d_2 = 0.10 \text{ m}$ , $h_3 = 2.1 \text{ m}$         | Torricelli; continuity; walking rule                                       | $Q \approx 0.0678 \text{ m}^3/\text{s}$ ; $h \approx 0.557 \text{ m}$               |

| #  | Problem type                                                                                     | Key equations                                            | Answer                                                                                             |
|----|--------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 14 | Cavitation: $D_1 = 1$ in throat, $D_2 = 2$ in exit, $p_{\text{atm}} = 14.5$ psia, $p_v = 3$ psia | $p_{\text{atm}} - \rho g h = p_v$                        | $h \approx 1.77$ ft; increase $D_1$ ; decrease $D_2$                                               |
| 15 | Circular streamlines: water $r = 4$ in, $V = 0.9$ ft/s; air $r = 290$ ft, 180 mph                | $dp/dr = \rho V^2/r$                                     | $4.71$ lb/ft <sup>3</sup> ; $0.572$ lb/ft <sup>3</sup>                                             |
| 16 | Z-shaped gate, hinge at the free surface, depths $h$ , $2h$ , $3h$ ; no force at stop            | $\Sigma M$ about A                                       | $b = \sqrt{54} h$                                                                                  |
| 17 | Air pipe $D = 20, 15, 10$ mm; water manometer $h = 1$ mm                                         | $(\rho_w - \rho_a)gh = \frac{1}{2}\rho_a(V_2^2 - V_1^2)$ | $V_3 \approx 11.0$ m/s; $p_1 < p_2$ false; $p_A < p_B$ true; $p_A < p_3$ either; $p_B < p_3$ false |

### Concept quiz answers (hydrostatics quadrant lab)

- Procedure: start with a full tank, drain until balanced for each weight, stop when  $h \approx R_2 - R_1$ .
- Largest balancing weight occurs at the highest water level.
- The quadrant's own weight is balanced out by the counterbalance.
- Pressure acts perpendicular to the surface; on the curved face it passes through the pivot and makes no moment.
- $h = f(W)$  comes from the sum of moments about the pivot:  $W \cdot \ell = F \cdot yR$  on the flat vertical face.